

Individual Plan

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1 Project Information

- Title: Surrogate Gradients for Gradient-Based Parameter Estimation in Simplified Neuron Models
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- Examiner: Eric Fransén
- Supervisor: Alexander Kozlov
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- Keywords: Automatic Gradient Descent, Adaptive Exponential Integrate-and-Fire Model, Simplified Neuron Model, Surrogate Gradients, Parameter Estimation

2 Background & Objective

This project is conducted within computational neuroscience, specifically focusing on parameter estimation for simplified biophysical neuron models. Large-scale brain simulations require efficient methods for fitting neuron model parameters to experimental data. While sophisticated Hodgkin-Huxley-type models can precisely capture membrane behavior, they require enormous computational resources. On the other hand, simplified neuron models (like the Adaptive Exponential Integrate-and-Fire Model, AdEx) offer drastically lower computational cost while reproducing diverse firing patterns faithfully. This makes them prime candidates for whole-brain simulations.

The project connects to recent developments in gradient-based optimization for neuron models. The Jaxley framework demonstrated success using automatic differentiation for HH-type models. However, it lacks support for simplified models due to their non-differentiable spike mechanism (the discrete reset when membrane potential crosses threshold). Surrogate gradients, originally developed for training spiking neural networks, offer a potential solution by replacing discontinuous derivatives with smooth approximations during backpropagation. This requires loss functions to be differentiable as well. Whereas non-gradient based parameter optimization strategies can rely on non-differentiable feature based loss functions, gradient-based differentiation requires differentiable loss functions. This opens up for the question on what loss function design is required to achieve successful parameter fittings.

To study loss function design, signal processing techniques might provide valuable insight. But mainly, this work bridges techniques from machine learning (surrogate gradients) with biophysical modeling, addressing a gap in current tools. It is of interest to computational neuroscientists working toward large-scale brain simulations where efficient and accurate parameter fitting is essential.

The project will continue and extend on the implementation and first results gained from the 'Project Course in High Performance Computing'.

3 Research Question & Method

The main question I want to answer is: "Can gradient-based parameter optimization produce useful parameter updates for the AdEx model?"

Related questions that will also be studied in this project are: (1) What loss function design is required to achieve successful gradient-based optimization for simplified neuron models? (2) How do different surrogate gradient functions affect optimization convergence and parameter quality? (3) How does gradient-based optimization compare to derivative-free methods in terms of accuracy and computational efficiency?

3.1 Objectives

The following milestones need to be completed in order to answer the research question:

- Implement AdEx model using a bio-physical neuron simulator that allows automatic differentiation.
- Extend AdEx implementation to allow for surrogate gradients.
- Design differentiable loss function that allows successful gradient based optimization.
- Implement testing framework that allows optimization on multiple experimental traces.
- Measure computational efficiency of optimization method
- Compare gradient-based strategie against feature based state-of-the-art genetic / non-differentiable alternatives.

3.2 Tasks

Some of the task have already been started within the HPC project course. For example, I will use my previously implemented AdEx model. The author identified these main tasks and their respective challenges:

- building a differentiable loss function — this will require careful study of existing approaches e.g. in signal processing or machine learning domains.
- hyperparameter tuning — wdentifying what influence is due to the non-convex paramter space, to the actual optimizers and to the loss function may be difficult.
- building the testing framework — this will be mostly an implementation overhead. I don't expect much difficulty, however, the main challenge may be the result of different data formats or signal lengths — handling time series data might be challenging.
- measuring computational efficiency — can be highly system and data dependent. Finding good testing metrics might be challenging.
- finding other state-of-the-art optimizers — will not be as easy since there is no standard benchmark set for parameter tuning in neuroscience. Not all code may be easily accessible as well.

3.3 Method

The methodology follows an experimental computational approach combining implementation, validation, and systematic evaluation:

1. Implementation within Jaxley Framework

The AdEx model will be implemented as a Jaxley channel using the JAX framework for automatic differentiation. Surrogate gradients will be implemented using JAX's `custom_vjp` functionality, which allows defining custom vector-Jacobian products. Three surrogate functions will be explored: sigmoid, exponential, and SuperSpike surrogates. This approach is appropriate because Jaxley has demonstrated success for HH-type models, and extending it to simplified models maintains compatibility with existing neuroscience workflows.

2. Implementation Verification

The AdEx implementation will be validated against Brian2, a widely-used reference simulator. Verification metrics include visual comparison of voltage traces, spike count matching, and spike timing differences (target: ± 1 ms). This cross-validation approach ensures implementation correctness before proceeding to optimization experiments.

3. Loss Function Development

Three loss function approaches will be investigated:

- MSE-based and absolute-error-based loss on raw voltage traces (baseline)
- Feature-based loss adapted from Guarino et al. [9], capturing spike timing, interspike intervals, firing rate, and subthreshold behavior
- Test (smooth) dynamic time warping as loss function alternative.

The feature-based approach requires developing differentiable approximations of discrete spike detection operations using soft thresholds and weighted averages. This is necessary because gradient-based optimization requires end-to-end differentiability.

4. Systematic Evaluation:

Parameter optimization will be evaluated using intracellular recordings from striatal projection neurons. Performance will be assessed through (1) visual comparison of fitted vs. experimental traces and (2) coincidence factor Γ for spike timing precision (benchmark: $\Gamma \approx 0.5$) and (3) computational cost measurements.

5. Comparative Analysis:

Gradient-based results will be compared against derivative-free baselines (genetic algorithms, grid search) to assess relative performance and computational efficiency.

This methodology is appropriate because it follows a systematic progression from implementation to validation to evaluation, allowing identification of specific failure modes and enabling iterative improvement of the loss function design.

3.4 Ethics and Sustainability

Ethics: This project uses publicly available experimental data from prior research [1]. No new animal experiments are conducted. The computational methods developed could ultimately reduce the need for animal experiments by enabling better in-silico modeling and prediction of neural behavior.

Sustainability: The project addresses computational sustainability in neuroscience. Simplified models like AdEx require orders of magnitude less computational resources than detailed HH-type models. Enabling efficient parameter fitting for these models could significantly reduce the energy consumption required for large-scale brain simulations. Additionally, gradient-based optimization, if successful, would be more computationally efficient than evolutionary algorithms that require many more simulation evaluations. No significant ethical concerns are raised by this work.

3.5 Limitations

The following are explicitly out of scope for this project:

1. **Multi-compartment models:** Only single-compartment AdEx neurons will be considered. Extension to complex morphologies (dendrites, axons) is not included.
2. **Network-level optimization:** The project focuses on single-neuron parameter fitting. Fitting parameters for networks of neurons is not addressed.
3. **All neuron types:** Evaluation will focus on striatal projection neurons. Comprehensive validation across diverse neuron types (cortical, cerebellar, etc.) is limited by available data and time.

4. **Hyperparameter optimization:** While some hyperparameter exploration will be conducted, exhaustive optimization of surrogate gradient parameters, learning rates, and loss function weights is not the primary focus.
5. **Real-time or embedded applications:** The implementation targets research use in Python/JAX, not deployment in real-time systems.
6. **Comparison with existing methods:** Only selected baseline methods (Nelder-mead, grid search) will be compared. Comprehensive benchmarking against all published optimization approaches is not feasible.
7. **Theoretical analysis:** The focus is empirical. Formal convergence proofs or theoretical analysis of surrogate gradient approximation quality are not included.

3.6 Risks

Risk	Impact	Likelihood	Mitigation
Gradient-based optimization fails to produce useful parameters	High	Medium	The preliminary work already shows this is challenging. If optimization consistently fails, the project will pivot to documenting failure modes and characterizing the loss landscape, which is itself a valuable contribution.
Non-convex loss landscape prevents convergence	Medium	High	Implement multiple initialization strategies, learning rate schedules, and gradient clipping. Explore different surrogate gradient functions and hyperparameters. Document which conditions enable vs. prevent convergence.
Differentiable feature extraction introduces bias	Medium	Medium	Validate soft feature approximations against hard (non-differentiable) counterparts. Quantify approximation errors and their impact on optimization.
Limited experimental data availability	Medium	Low	The Johansson and Silberberg dataset [1] is already available. If additional data is needed, publicly available datasets can be used.
Computational resource constraints	Low	Low	The project uses simplified models specifically because they are computationally efficient. Standard laptop hardware is sufficient for most experiments. EECS Cluster can also be used as backup and for computations using GPUs.

Table 1: Risk Assessment

Contingency plan: If gradient-based optimization proves fundamentally unsuitable for AdEx parameter estimation, the project can still contribute by (1) documenting why this approach fails, (2) characterizing loss function requirements, and (3) providing a validated, differentiable AdEx implementation for future research.

4 Evaluation & News Value

The project objectives will be evaluated using the following criteria:

4.1 Quantitative Measures

1. **Coincidence Factor (Γ):** The primary metric for parameter fitting quality. Success threshold: $\Gamma > 0.5$, matching or approaching the $\Gamma \approx 0.82$ – 0.83 achieved by Jolivet et al. [2] for well-fitted AdEx models.
2. **Spike count accuracy:** Percentage of fitted models producing the correct number of spikes (± 1 spike tolerance).
3. **First spike timing error:** Mean absolute error in predicting the first spike time (target: $< 5\text{ms}$).
4. **Convergence rate:** Number of optimization iterations required to reach a stable loss value.
5. **Computational efficiency:** Wall-clock time per optimization run compared to genetic algorithm baselines.

4.2 Qualitative Measures

1. **Loss function characterization:** Documentation of which loss function designs enable vs. prevent successful optimization.
2. **Surrogate gradient comparison:** Qualitative assessment of how different surrogate functions affect optimization behavior.
3. **Failure mode analysis:** Identification and documentation of systematic failure patterns (e.g., convergence to non-spiking solutions).

4.3 Success Criteria

- **Full success:** Gradient-based optimization achieves $\Gamma > 0.5$ with computational advantage over baselines.
- **Partial success:** Gradient-based optimization produces spiking behavior with correct spike count but limited timing precision.
- **Documented failure:** Clear characterization of why gradient-based optimization fails, with recommendations for future work.

Any of these outcomes constitutes a valid scientific contribution.

4.4 Scientific Relevance

This work addresses a methodological gap in computational neuroscience: the lack of gradient-based parameter fitting tools for simplified neuron models. While Jaxley has demonstrated success for HH-type models, simplified models like AdEx remain limited to derivative-free optimization methods despite their importance for large-scale brain simulations.

4.5 Innovation/News Value

1. **First systematic study** of loss function requirements for gradient-based optimization of simplified neuron models in biophysical simulation contexts.
2. **Differentiable AdEx implementation** in Jaxley, enabling the broader community to use gradient-based methods for simplified models.
3. **Bridge between SNN training and biophysical fitting:** Demonstrates whether techniques from machine learning (surrogate gradients) can be transferred to neuroscience applications.
4. **Practical guidance** on loss function design for researchers attempting similar approaches.

4.6 Target Audience

1. Computational neuroscientists working on large-scale brain simulations who need efficient parameter fitting methods.
2. Developers of neural simulation frameworks (Jaxley, Brian2, NEST) interested in extending automatic differentiation support.
3. Machine learning researchers working on spiking neural networks who may benefit from insights on loss function design.
4. Researchers working toward whole-brain simulation using simplified models.

5 Pre-study

5.1 Literature Study Focus Areas

1. **Simplified neuron models:** Mathematical formulation, parameter interpretation, and fitting approaches for AdEx and related models [3], [4].
2. **Surrogate gradient methods:** Theory and practice of surrogate gradients for spiking neural networks [5], [6], [7]. Recent work by Yu et al. [8] demonstrates that surrogate gradients can enable learning from temporal spike patterns beyond firing rates, which is directly relevant to whether gradient-based optimization can capture spike timing precision.
3. **Automatic differentiation for neuroscience:** Jaxley framework and related approaches [9], [10].
4. **Parameter optimization for neuron models:** Evolutionary algorithms, grid search, and other derivative-free methods [11], [12].
5. **Loss function design:** Feature-based losses, spike train metrics, and differentiable approximations [2], [12].
6. **Differentiable time series alignment:** Soft-DTW [13] provides a differentiable version of dynamic time warping that enables gradient-based optimization for time series comparison—potentially applicable as a loss function for voltage trace fitting that is robust to small temporal misalignments.
7. **Time warping for neural data:** Methods for temporal alignment in neural recordings [14], [15] address the problem of trial-to-trial timing variability in neural responses. Williams et al. [14] developed robust time warping methods for large-scale neural recordings, while Lawlor et al. [15] combined time warping with neural encoding models. These approaches may inform loss function design that accounts for temporal variability.
8. **Spike train distance metrics:** Classical metrics for comparing spike trains, including the van Rossum metric [16], provide mathematically grounded ways to quantify spike train similarity. Understanding these metrics may guide the design of differentiable approximations suitable for gradient-based optimization.

5.2 Knowledge Acquisition

- Review of primary literature in computational neuroscience and machine learning
- Study of Jaxley and JAX documentation and source code
- Analysis of existing AdEx parameter fitting implementations

5.3 Preliminarily Important References

- Deistler et al. (2025) [9] — Jaxley framework
- Neftci et al. (2019) [5] — Surrogate gradient learning review
- Brette & Gerstner (2005) [3] — Original AdEx paper
- Guarino et al. (2025) [12] — Feature-based loss function
- Jolivet et al. (2008) [2] — Coincidence factor metric
- Zenke & Ganguli (2018) [6] — SuperSpike surrogate
- Cuturi & Blondel (2017) [13] — Soft-DTW differentiable loss
- Yu et al. (2025) [8] — Surrogate gradients for spike timing learning
- Williams et al. (2020) [14] — Time warping for neural recordings
- Lawlor et al. (2018) [15] — Time-warp-Poisson neural models
- Houghton (2009) [16] — Van Rossum spike train metric

6 Conditions & Schedule

Resources Required:

- Hardware: Personal laptop (Apple M1 Silicon) - sufficient for most experiments
- Server: Access to EECS Cluster — necessary for GPU runtime tests
- Software: Python 3.14+, JAX, Jaxley 0.12+, standard scientific Python stack
- Data: Data: Intracellular recordings from Johansson Silberberg [1] — publicly available
- Code: Existing implementation from HPC project course (AdEx model, surrogate gradients)

Supervisor Involvement:

Alexander Kozlov will provide guidance on:

- Neuroscience domain knowledge and biological interpretation
- Connection to ongoing striatum modeling work
- Feedback on experimental design and results interpretation

Regular meetings (approximately bi-weekly) will be scheduled to discuss progress and address challenges.

6.1 Timeline

See table 2.

7 Use of Generative AI

Generative AI (Claude, Anthropic) was used in the preparation of this individual plan for:

1. Document formatting: Assistance in organizing and formatting the plan according to the template requirements.
2. Writing refinement: Improving clarity and conciseness of technical and non-technical descriptions.

All content was reviewed and validated by the author. The scientific content, research questions, and methodology are the author’s own work, building on preliminary results from the HPC project course.

References

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Month	Week	Days	Pilot testing	Main step	Substeps	Deadlines	What I should have decided	Thesis writing
February	6	2–6		Literature review & pre-study	Complete individual plan. Refine research questions.	Send individual plan to supervisor.		Related work
	7	9–13			Write background section. Review surrogate gradient literature.	Get OK on individual plan from examiner.		Related work
	8	16–20	Test different surrogate gradient functions	Loss function development	Implement differentiable feature extraction. Test MSE baseline.		Which surrogate gradient function to use as default.	Background
	9	23–27			Implement feature-based loss. Test DTW alternatives.	Loss function v1 complete.	Loss function architecture decision.	
March	10	2–6	Parameter sensitivity analysis	Optimization experiments	Single trace optimization. Hyperparameter tuning.			Background & Method
	11	9–13			Document convergence behavior. Analyze failure modes.	Initial results available.	Optimal learning rate and surrogate slope.	
	12	16–20	Test on different neuron recordings		Build multi-trace testing framework.			Method
	13	23–27			Evaluate on multiple experimental traces.	Testing framework complete.	Evaluation protocol finalized.	
	14	30–3		Break / buffer				
April	15	6–10	Compare GA implementations	Baseline comparisons	Implement genetic algorithm baseline.			Results
	16	13–17			Run comparative experiments. Document accuracy differences.	Comparative analysis complete.	Which baseline methods to include.	
	17	20–24		Efficiency measurements	Measure wall-clock time. Profile GPU vs CPU.			Discussion
	18	27–1			Convergence rate analysis.			
May	19	4–8			Final experiments.	All experiments complete.		Results section
	20	11–15		Thesis writing	Complete results section.			Working on all parts
	21	18–22			Complete discussion and conclusion.	Draft complete. Send to supervisor.		
	22	25–29		Break / buffer				
June	23	1–5		Revision & submission	Address feedback. Final revisions.	Send final version to examiner.		Final revision
	24	8–12			Prepare presentation.	Thesis submitted.		

Table 2: Project Timeline